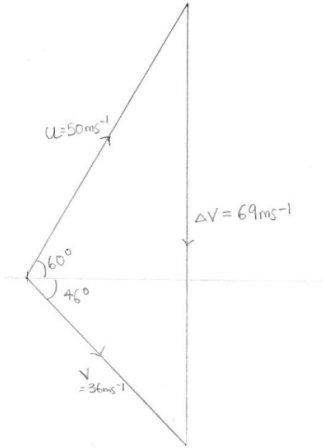


2	
(a)	$u_x = \frac{220}{8.8} = 25 \text{ m s}^{-1}$
	$u_y = 43.164 \text{ m s}^{-1}$
	$\tan \theta = \frac{43.164}{25}$ $\theta = 59.9^\circ (1dp)$
	$u = 50 \text{ m s}^{-1} (2sf)$
(b)(i)	change in velocity, Δv $= 69 \text{ m s}^{-1} (2sf)$
	 The diagram illustrates the vector subtraction $\Delta v = u - v$. Vector u has a magnitude of 50 m s^{-1} and is directed horizontally. Vector v has a magnitude of 36 m s^{-1} and is directed at an angle of 46° below the horizontal. The resultant vector Δv, representing the change in velocity, has a magnitude of 69 m s^{-1} and is represented by the vertical line segment connecting the tip of v to the tip of u. The angle between u and v is marked as 60°.
(b)(ii)	magnitude of $v = (36 \pm 2) \text{ m s}^{-1}$ direction: $(46 \pm 2)^\circ$ below horizontal

